▶ NISAR SATELLITE

THE PLANET MAPPER

The NASA ISRO Synthetic Aperture Radar or the NISAR satellite, the largest collaboration in space between the US and India, is scheduled for launch from Sriharikota, Andhra Pradesh, on July 30 at 5.40 pm. One of the most sophisticated radar satellites till date, it has been in the works since 2014. Its goal is to track subtle changes on the Earth's surface that have hitherto proved virtually undetectable. To achieve this, it will scan the planet twice every 12 days in unprecedented detail, literally down to a centimetre, regardless of cloud cover, darkness or foliage.

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Graphic by **Tanmoy Chakraborty**

$1.2 BN
NASA budget. It makes NISAR one of the most expensive Earth-observing satellite missions undertaken

₹788 CRORE
ISRO's estimated work share cost
(Final budget has yet to be announced)

ENSURING PRECISION

Dual frequency radar

- NISAR will be the first satellite to observe the Earth with two kinds of radar—an L-band SAR (Synthetic Aperture Radar) from NASA's Jet Propulsion Laboratory and an S-band SAR built by Space Applications Centre in Ahmedabad
- Both will use a 12-metre unfurlable mesh reflector antenna
- The two different wavelengths will help capture richer data

1-centimetre scale

- Placed in a low orbit, NISAR will be able to detect minute changes on the Earth's surface
- The applications include sighting ground deformation, ice sheet movement and vegetation dynamics, sea ice classification, ship detection, shoreline monitoring, storm characterisation, changes in soil moisture, mapping and monitoring of surface water resources and disaster response, which can help predict floods, volcanic eruptions and even earthquakes

5 YEARS
Minimum duration of satellite operations

